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Evaluation of chloride and pesticide transport in a fractured clayey till using large undisturbed columns and numerical modeling

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Abstract. Saturated groundwater flow and tracer experiments using fluorescent dye, chloride, and the herbicides mecoprop and simazine were carried out in the laboratory using three large-diameter (0.5 m) undisturbed columns of fractured clayey till. Hydraulic conductivity of the columns ranged from 10^{-5} m/s in the shallowest column (1 m depth) to 10^{-7} m/s in the deepest column (4 m depth) and were similar to field-measured values for these deposits. Results of the tracer experiments are consistent with a conceptual model of advective transport along the fractures combined with diffusion into the fine-grained matrix between the fractures. Arrival of the chloride tracer in the effluent was highly retarded relative to fracture flow velocities calculated on the basis of the cubic law and measured values of fracture spacing and hydraulic conductivity. The herbicides were more strongly retarded than the chloride at low flow rates, but at higher flow rates the herbicides arrived with the chloride, indicating the influence of nonequilibrium sorption of the herbicides to fracture walls and the matrix solids. The columns were dismantled following the tracer experiments and mapping under UV light showed that nearly all of the visible, weathered fractures (and the few root holes in the case of the shallowest sample) were active transport pathways, with the dye appearing mainly on the fracture surfaces and as a "rim" in the adjacent matrix. Concentration profiles measured perpendicular to the fracture surfaces showed that the herbicides had also moved into the matrix, apparently by diffusion. Simulations of solute transport with a discrete fracture flow/matrix diffusion model showed that the simulations could be "fit" to the data if all of the visible fractures were hydraulically active, but could not be fit if all or most of the flow was channelled through just the primary fractures (defined by prominent oxidation stains). Simulations with an equivalent porous media (EPM) model could not fit the data using the measured total porosity as the effective porosity. The simulations could likely be fit with a smaller value of effective porosity, but this would limit applicability to field situations because fitted effective porosity is expected to change with physical scale and residence time of the solute in the soil.

1. Introduction

Recent field investigations [*D'Astous et al.*, 1989; *Jørgensen*, 1990; *Jørgensen and Fredericia*, 1992; *McKay et al.*, 1993a, b; *McKay and Fredericia*, 1995; *Jørgensen and Spliid*, 1998; *Jørgensen et al.*, 1998a] have shown that in fractured clayey glacial deposits dissolved contaminants or environmental tracers can migrate at faster rates than those measured in unfractured clayey glacial deposits [*Johnson et al.*, 1989; *Desaulniers et al.*, 1981]. The migration of dissolved contaminants in clayey glacial deposits is mainly controlled by advective transport through the fractures and biopores (such as burrows and root casts) combined with diffusion of solutes into the relatively immobile pore water in the clayey matrix between fractures. The diffusion process, which influences both reactive and nonreactive solutes, can attenuate solute migration relative to the velocity of flow in the fractures and is often referred to as matrix diffusion [*Freeze and Cherry*, 1979]. The migration of contaminants, such as pesticides, can also be influenced by chemical or biochemical decomposition and by sorption to soil materials. The attenuating effect of these reactions depends on the nature of the sediment, the contaminant and the geochemical environment.

In cases with high hydraulic gradients (0.1–1.0) such as those

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that could occur during seasonally high groundwater levels and rain storms, the flow velocity in fractures can be in the order of tens to hundreds of meters per day [*McKay et al.*, 1993b; *Jørgensen*, 1995; *Hinsby et al.*, 1996; *Jørgensen et al.*, 1998b]. At these high flow velocities there would likely be insufficient residence time for decomposition or for sorption of the contaminants to reach local equilibrium. Hence it is possible that contaminants could move rapidly with only small declines in concentration. In field experiments in clay-rich soils rapid leaching of pesticides, which were expected to be essentially immobile because of their high distribution coefficients (K_d values), have been observed and are believed to result from rapid transport in fractures and biopores [e.g., *Brown et al.*, 1995; *Traub-Eberhard et al.*, 1995].

Assessment and prediction of groundwater flow and contaminant transport in fractured or biopore dominated soils or clay-rich deposits have proven to be very uncertain because of difficulties in measuring critical parameters such as hydraulic conductivity, fracture/biopore aperture, and degree of interconnection. Field- and laboratory-measured values of hydraulic conductivity at several fractured clay sites in Canada and Denmark have shown varying degrees of sensitivity to scale of measurement and disturbance, particularly borehole smearing, which closes fractures and reduces field-measured conductivity values [*Hendry*, 1982; *Keller et al.*, 1986; *D'Astous et al.*, 1989; *Fredericia*, 1990, *McKay et al.*, 1993a]. Even in cases where efforts were taken to minimize smearing in the boreholes it was found that there was still uncertainty in hydraulic conductivity values because of the influence of the physical scale of measurement. Values measured using small-scale methods can vary greatly depending on whether they intersect fractures or matrix. Values measured using large-scale methods are expected to show less variability but are incapable of distinguishing between the influence of many small fractures and a few highly conductive fractures.

Numerical modeling of flow and transport in fractured clayey deposits can be a useful investigative tool, but application for predictive purposes is greatly limited by difficulties in measuring critical parameters (as previously described) and by the scarcity of controlled field or laboratory experiments needed for calibration of models. The basic numerical modeling approaches used for flow and solute transport in fractured media are (1) discrete fractures (DF) with no matrix interaction, (2) discrete fractures with advection and/or diffusion into the matrix blocks (DFMD) between fractures, (3) single porosity continuum or equivalent porous medium (EPM) approaches, and (4) dual porosity continuum (DP) approaches with transfer between pore regions. In the DF approach (described by *Smith and Schwartz* [1984]) flow and solute transport is assumed to occur only in explicitly defined fractures. Attachment to fracture walls can be allowed, but there is no interaction between the fracture pore water and the matrix between fractures. This approach may be appropriate for very low porosity rocks, but experimental investigations in fractured high-porosity deposits (such as clay tills) have shown that diffusion into the matrix is a major factor in controlling solute transport [*Grisak et al.*, 1980; *McKay et al.*, 1993b; *Hinsby et al.*, 1996]. Application of DF models to these high porosity materials tends to greatly overestimate contaminant transport rates. Models that incorporate both transport through discrete fractures and diffusion into the matrix (DFMD models) have shown that fractures can greatly increase transport rates relative to unfractured clay or rock, but rates are still orders of magnitude slower than would be predicted by the DF models without matrix diffusion [*Sudicky and McLaren*, 1992; *Harrison et al.*, 1992].

Both the DF and the DFMD approaches are dependent on estimates of fracture aperture and fracture geometry (spacing, length, and orientation). Fracture geometry can often be measured in core samples and outcrop, but it is usually not possible to directly measure aperture or distinguish between fractures that are "open" to flow and those that are "closed" under the in situ stress conditions. Aperture values are generally calculated using the so called cubic equation with measured hydraulic conductivity and fracture spacing/orientation data [*Snow*, 1969; *McKay et al.*, 1993a]. This calculation assumes that all fractures have the same value of aperture, that there is no flow in the biopores (or they are represented as part of the fracture system), and there is no channeling of flow along larger aperture pathways within individual fractures. These assumptions introduce significant uncertainty into calculated apertures and subsequent transport simulations. In cases where most of the flow occurs through a few larger aperture fractures, transport could be much faster than predicted by simulations assuming flow is evenly distributed between fractures.

In the equivalent porous medium (EPM) approach the fractured system is represented by a single porosity continuum (or series of continuums for heterogeneous materials). Flow and transport can be represented by plug flow or by the advection dispersion equation [*Freeze and Cherry*, 1979; *Berkowitz et al.*, 1988]. This approach is numerically simple and is widely used. The EPM approach is valid only in systems where the flow and transport properties of the fracture network do not vary greatly with scale or position, so that a representative elemental volume (REV) for the material can be defined at a physical scale that is substantially smaller than the system being modeled. With respect to flow, conditions that favor EPM behavior include high fracture density, frequent fracture intersections, and uniform aperture size [*Long et al.*, 1982]. If a contaminant diffuses rapidly into the matrix blocks, relative to the rate of advection along the fractures and if the spacing between fractures is small, then the system can be modeled using the advection dispersion equation with effective porosity equal to total porosity [*van der Kamp*, 1992]. In cases with larger fracture spacing or slower diffusion the EPM approach may still "fit" the data but do so at an effective porosity less than the total porosity, and this fitted parameter will change with the physical and temporal scale of the transport problem. As a result, the EPM approach is most appropriate for cases where there is no significant matrix porosity (in which case effective porosity equals fracture porosity) and for cases where the material is highly fractured and diffusion is rapid (in which case effective porosity equals total porosity).

The dual porosity (DP) continuum approach [*Huyakorn et al.*, 1983] has the same REV limitations as the EPM approach, but like the DFMD approach it has the added advantage that transfer between the fractures and matrix can be included. There are a variety of methods of accounting for transfer between pore regions, ranging from "fitted" transfer coefficients to measured diffusion coefficients and ped sizes. This approach has been used in highly fractured soils but is expected to be less appropriate for clayey tills where fractures are often more widely spaced.

The preceding modeling approaches can be used in either a deterministic or a stochastic mode. In the commonly used deterministic approach a single set of input values, often cho-

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sen on the basis of measured or assumed parameter values and boundary conditions, leads to a solution consisting of a concentration versus time breakthrough curve at a specific point within a system or the distribution of concentration values within the system at a given time. If concentration data from experiments or monitoring of existing contaminant plumes is available, the model may be fit to the data by varying one or more of the input parameters. The reliability of the solution is dependent on the uncertainty of the input data, which is not explicitly stated. In a stochastic approach, uncertainty of parameter values (such as fracture aperture) can be included with the input data and results in a probabilistic solution [*Fetter*, 1993]. Although the stochastic approach has some advantages, it is also limited by the difficulties related to measurement of parameters such as fracture aperture and by the scarcity of case studies for model calibration.

There are only a few cases where numerical models have been applied to solute transport data from fractured, clay-rich deposits. *Pankow et al.* [1986] showed that in one case in highly fractured unconsolidated silts the EPM approach was appropriate, but in another case, in sparsely fractured shales, the EPM approach was not successful. In a field tracer experiment in clay-rich glacial deposits *McKay et al.* [1993b] showed that both the EPM and the DFMD approaches could describe transport in the highly fractured (spacing 0.01–0.1 m) upper 3 m of the deposit, but below 3 m depth, where fractures were widely spaced, transport followed a few individual fractures and the EPM approach was not appropriate. In a recent study of a 16-year-old tritium plume in a weathered and fractured shale (typical fracture spacing estimated at <0.15 m) the resulting plume had a smooth shape and could be fit with either an EPM [*McKay et al.*, 1997] or a DFMD [*Stafford et al.*, 1998] approach. In this case, where fracture geometry and aperture were not well characterized, the EPM approach was preferred because it fit the plume with a unique set of parameters, while there were a variety of fracture scenarios that could have fit the same plume with the DFMD approach.

The objectives of the investigations presented in this paper are to (1) experimentally investigate groundwater flow and transport of reactive and nonreactive solutes in undisturbed columns of clayey till which are large enough to contain a significant number of fractures and biopores; (2) apply and compare two types of models, single-continuum (or EPM) and discrete fracture/matrix diffusion (DFMD), for simulating contaminant transport in the samples; and (3) obtain fracture scale data and modeling parameters, particularly fracture spacing, hydraulic aperture and fracture porosity, for future use in field-scale modeling and contaminant risk assessment.

2. Field Site Description

The till columns used in the experiments were collected in an orchard near the village of Havdrup, 40 km southwest of Copenhagen, in Denmark (Figure 1). The site is located on a slightly undulating Weichselian till plain approximately 15–16 m thick. Based on information from wells and excavations at the site there is a high degree of lateral uniformity in the glacial sequence. This is illustrated by a 0.5- to 1.0-m-thick sandy layer which occurs over the entire site (approximately 2 ha) and is also found in wells adjacent to the site. The upper 3 m of the till is oxidized to a yellowish brown color and is highly fractured. Near-vertical fractures with prominent staining (primary fractures) are found throughout the oxidized zone and in some cases extend several meters down into the grey unoxidized clays. In the unoxidized zone staining is found mainly along the primary fractures. Spacing of these fractures increases from 5–10 cm at a depth of 1.5 m to 0.5–1.5 m at a depth of 2.5–3.0 m. Below 3 m depth fractures often occur in groups which can be widely separated (>10 m) by apparently unfractured clayey matrix. The deep fractures are primarily vertical and parallel to one another with visible oxidation staining to approximately 4 m depth. At greater depth unstained fractures are indentified by the presence of surfaces with slickensides [*Jørgensen and Spliid*, 1998].

Figure 1. Denmark, with field site location indicated.

In the upper 1.5 m of the oxidized zone, faintly stained secondary fractures (soil ped interfaces) with a wide range of orientations were present. Spacing of these fractures are 1.5–3.0 cm at 1 m depth. Numerous biopores (burrows from roots and small animals) occur in the upper half meter of this zone. Frequency of biopores decreases with depth, and these are absent below a depth of approximately of 1.5–2.0 m. Most of the primary and secondary fractures in the upper few meters appear to be caused by dessication during periods of low water table. The widely spaced vertical fractures in the unoxidized zone are likely due to glacial tectonic disturbance. The glacial sequence rests upon sandy limestone and marl from the upper Paleocene which is underlain by whitish Danien bryozoan limestone. The two types of limestone constitute the main aquifer of the region and are heavily used for the water supply of Copenhagen.

Field investigations by *Jørgensen and Fredericia* [1992] suggest that groundwater flow in the till deposits is most intensive in the upper 10 m of the sequence. The position of the water table varies from near ground surface in winter and spring to depths of 3 m in summer [*Jørgensen*, 1995]. Hydraulic conductivity of the unfractured clayey till matrix in typical Danish clayey till is very low (10^{-9}–10^{-11} m/s) [*Fredericia*, 1990; *Foged and Wille*, 1992], so it is assumed that groundwater infiltration and transport of contaminants in this deposit is controlled mainly by the observed fractures and biopores. This is sup-

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Table 1. Physical and Textural Properties of Column Material

Column	Depth, m	Clay, Silt, Sand, and Gravel, %	Matrix Porosity, %	Fracture Spacing, m	
				Primary Fractures	Secondary Fractures
I	1.0–1.5	18, 28, 50, 4	32	0.06	0.025
II	2.0–2.5	21, 24, 51, 4	31	0.10	···
III	4.0–4.5	21, 29, 48, 2	25	0.16	···

ported by the presence of tritium and the herbicides atrazine and simazine in pore water samples to the maximum bore depth of 8 m in the till [*Jørgensen and Spliid*, 1998].

3. Materials and Methods

3.1. Excavation and Sampling of Undisturbed Till Columns

Collection of undisturbed till columns was carried out in a 6-m-deep excavation in the orchard during moist weather conditions in autumn. The columns, each 0.5 m in diameter by 0.5 m high, were collected at depths of 1–1.5, 2–2.5 (both in yellowish brown oxidized till), and 4–4.5 m (grey reduced till), hereafter referred to as columns I, II, and III, respectively. The two shallowest samples (columns I and II) contained numerous fractures and appeared to be representative of the till at their respective depths. The deepest sample (column III) contained three prominent fractures and appeared to be representative of the widely spaced (often >10 m) groups of fractures found in the unoxidized till. Fracture spacing, grain size analyses for the bulk column material, and matrix porosity values, θ_m (European Testing Committee, unpublished report, 1995) are shown in Table 1. No textural changes were observed in or along the fractures, suggesting that fracture infillings are minimal. Further geological and mineralogical description of the material is given by *Jørgensen and Fredericia* [1992].

Excavation of the columns was done with hand tools to prevent disturbance from machines. After the columns were excavated but before being disconnected at their base from the geological formation, they were covered with a flexible rubber membrane and fluid rubber was injected between the membrane and the sample. Before hardening, the fluid rubber enters a few millimeters of the till matrix and seals the outer surface of the columns, eliminating problems with flow along this boundary during the experiments. The sample was enclosed in a temporary steel casing and the two ends of the column were capped with a 1-cm-thick quartz sand layer and a stainless steel end plate. The steel-encased columns were then transported to the laboratory. Further description of the sampling method is given by *Jørgensen and Foged* [1994].

3.2. Column Experiments

Realistic in situ conditions for the laboratory experiments were achieved by installing the columns in a flexible wall triaxial setup with confining pressure of 40 kPa and temperature of 10°C. During setup in the laboratory stability of the columns were ensured by applying a vacuum of −30 kPa to the inside of the columns after having removed the steel transport casing. An outline of the experimental setup is shown in Figure 2. Detailed description of installation procedure and the experimental setup is given by *Jørgensen and Foged* [1994].

Bulk hydraulic conductivity values for the saturated columns were measured by recording effluent yield from the columns at different hydraulic gradients. Hydraulic gradient was controlled by the elevation difference between a constant head influent reservoir and the end of the effluent tubing. Each hydraulic gradient step was maintained for a period of several hours to several days, and flow was monitored to ensure that steady state conditions had been reached.

Experiments with transport of a nonreactive solute were performed by leaching the saturated columns with a ground water solution containing a $CaCl_2$ tracer followed by flushing with clean groundwater. An influent concentration of $CaCl_2$ equivalent to 3000 mg Cl^-/L was applied in column I and II. The influent concentration of Cl^- for column 3 was 1700 mg/L. Pesticide transport experiments were performed with the herbicides mecoprop (MCPP) and simazine. Both pesticides were formulated commercial products: M-propionat 50 (MCPP) and DLG simazin, both manufactured by Esbjerg Kemi A/S, Denmark. Influent concentration of MCPP was 54.7 mg active ingredient/L (a.i./L) in column I and 57.6 mg a.i./L in columns II and III. The influent concentration of simazine was 4.8 mg a.i./L for columns II and III. A fluorescent dye (Na-fluorescein, input concentration of 2000 mg/L) was passed through each column at the end of the solute transport experiments, and the columns were dismantled and examined under UV light to identify hydraulically active fractures and biopores.

The water used for influent in the experiments was obtained from a confined aquifer at a site that is geologically similar to the site from which the columns were taken. The chemical composition (major ions in mg/L) of the water in the aquifer was TDS, 730; Ca^{2+}, 120, Mg^{2+}, 31; Na^+, 31; HCO_3^-, 408; SO_4^{-2}, 73; and Cl^-, 72; the pH was 7.4. All experiments were carried out using unit hydraulic gradient. Flow rates were monitored throughout the experiments.

3.3. Chemical Analyses

Chloride concentrations in the influent and effluent were continuously monitored with an electric conductivity meter (Radiometer A/S) and data logger. A test performed [*Hinsby et al.*, 1996] showed that the conductivity meter measurements could be directly correlated with the breakthrough of chloride (measured by specific analyses). Recorded effluent conductivities (e_e) were normalized (e^*) to influent conductivities (e_i) with the expression $e^* = (e_e - e_{\text{background}})/(e_i - e_{\text{background}})$. Influent and effluent pH were measured using a Radiometer pH meter. Extraction of pesticides in water samples was made with dichloromethane. Monitoring of simazine and mecoprop in the influent and effluent (percolate) extracts was carried out with high-performance liquid chromatography (HPLC) and UV detection at 229 nm. Effluent concentrations of pesticides (C_e) were normalized (C^*) to influent concentration (C_i) as $C^* = C_e/C_i$ because $C_{\text{background}}$ of pesticides was below detection limit (0.01 μgram/L). Fracture wall matrix materials were sampled from the columns after the pesticide experiment. Soil solutions from these samples were extracted with dichloromethane and the extracts were subject to HPLC analysis as described above.

4. Experimental Results and Discussion

4.1. Bulk Hydraulic Conductivity and Flow Rates

The results of the saturated hydraulic conductivity tests carried out prior to the tracer experiments are shown in Figure 3, which presents flow rate versus hydraulic gradient for the three

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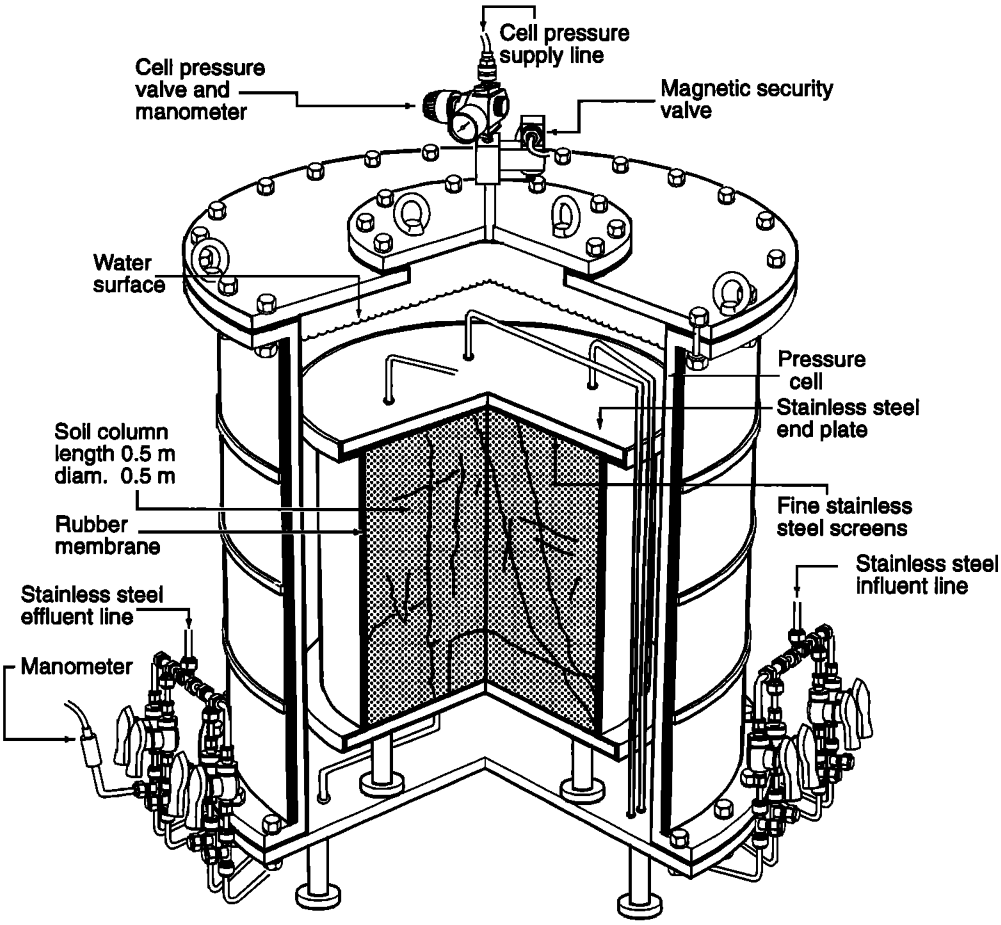

Figure 2. Flexible wall permeameter setup for flow and contaminant transport experiments using large undisturbed soil columns.

columns. The straight line relationship between flow and gradient for each column indicates that the columns are physically stable; that is, fracture openings (apertures) and contacts between columns and rubber membranes were not influenced by changes in the effective stress over the range of the hydraulic gradients applied. The hydraulic conductivity (K) value for each column was determined by fitting a regression line to the data on Figure 3 using the Darcy equation:

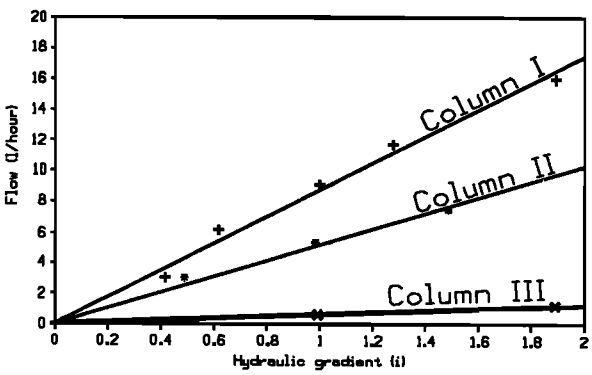

Figure 3. Effluent flow versus applied hydraulic gradients for constant head permeameter tests carried out prior to the tracer experiments. Applied permeameter confining pressure was 60 kPa.

$$K = \frac{Q}{Ai} \tag{1}$$

where

K hydraulic conductivity (m/s);
A area of outflow (m^2);
Q outflow rate (m^3/s);
i hydraulic gradient (nondimensional);
Q/i slope of regression line on Figure 3 (m^3/s).

Conductivity values decrease significantly with depth from approximately 10^{-5} m/s (equivalent to medium-fine sand) at 1.5 m depth to 10^{-7} m/s at 4–4.5 m depth in the profile (Table 2). The decrease in K values with depth inversely correlates to the increase in spacing of fractures and the disappearance of root holes (present only in column 1) observed in the laboratory columns and in the excavations. The lab-measured saturated hydraulic conductivities (K) are similar to field-measured bulk values in Danish clayey tills [*Fredericia*, 1990]. These values of K are several orders of magnitude higher than typical values measured in unfractured samples of clayey till. In a clayey till similar to the till of this investigation, the hydraulic conductivity of the till matrix was 2 to 7 × 10^{-10} m/s [*Foged and Wille*, 1992], which is much lower than the values measured for the columns. On the basis of these figures the flow in the till matrix of the columns was estimated to be 10^{-3} to 10^{-5} times that of the bulk flow in the columns.

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Table 2. Flow Rates and Hydraulic Conductivity Values

		Flow Rates,* m/d		Hydraulic Conductivity, m/s		
Column	Depth, m	$CaCl_2$ Experiment	Pesticide Experiment	Prior to Tracer Experiments	$CaCl_2$ Experiment	Pesticide Experiment
I	1.0–1.5	1.04	1.52	1.2×10^{-5}	1.2×10^{-5}	1.8×10^{-5}
II	2.0–2.5	0.588	0.234	4.4×10^{-6}	6.9×10^{-6}	2.7×10^{-6}
III	4.0–4.5	0.053	0.012	2.3×10^{-7}	6.1×10^{-7}	1.4×10^{-7}

*Expressed as average volumetric flow rate divided by cross-sectional area of column.

Flow rates in the individual columns varied by factors of 1.5–4.4 during the subsequent tracer experiments, as shown in Figure 4 and Table 2. This occurred even though the hydraulic gradient remained constant, at a value of 1. It is believed to be due primarily to accumulation of gas from biologic activity in fractures and macropores. Gas bubbles were observed intermittently in the effluent lines during all of the tracer experiments. These changes were not observed during the constant gradient steps of the preceding flow tests, but this may have been due to the shorter duration of the flow tests. Variations in flow rate attributed to biogenic gas production also occurred in saturated flow tracer experiments in columns of fractured till from another nearby site in Denmark [*Hinsby et al.*, 1996]. Hydraulic conductivity values calculated for each tracer experiment, along with the values calculated prior to the tracer experiments, are shown in Table 2.

4.2. Chloride and Pesticide Transport Experiments

Normalized effluent breakthrough curves for chloride (measured as electric conductivity) for each column are shown in Figure 5. The slope of the breakthrough curves tends to decrease after an initial rapid rise in concentration. Rapid advection of solutes through fractures provides the characteristic early breakthrough occurring at much less than 1 pore volume (calculated on the assumption that total porosity is approximately equal to matrix porosity). The subsequent flattening of the curves is due to diffusion of the solute from the fractures into the porous matrix and is similar to what has been observed in other solute tracer experiments in fractured clay-rich materials [*McKay et al.*, 1993b; *Jørgensen*, 1995; *Hinsby et al.*, 1996]. In each case the effluent concentration reached approximately 80% of the influent value by the end of the chloride injection. After the start of flushing there was a rapid drop in concentration followed by slowly decreasing concentrations due to continuing diffusion of tracer out of the matrix and back into the fractures. The chloride is relatively nonreactive in the clays, and it is expected that it would eventually all be flushed out, although because of matrix diffusion this could take a very long time.

Figure 6 shows the normalized effluent breakthrough of a pesticide pulse (simazine and mecoprop) applied to each of the columns. In all three experiments the first arrival of the pesticides occurs very rapidly, followed by peak concentrations occuring within 0.14–0.49 pore volumes. Mass balance data for the pesticide experiments are shown in Table 3. By flushing the column with approximately 2 pore volumes (PV) of pesticide-free solution, 75–91% and 21–93% of the applied MCPP and simazine, respectively, were leached. For both pesticides the lowest values of peak concentrations and percentage of recovery of pesticides in the effluent were measured in column 3, which also had the slowest arrival of pesticide peaks (0.43–0.49 PV). In columns I and II, which had higher flow rates, peak concentrations (as C/C_0) of simazine and MCPP were both in the range of 0.45–0.56, which was much higher than that observed in column III, where peak concentrations of 0.07 for simazine and 0.26 for MCPP were measured. The greater at-

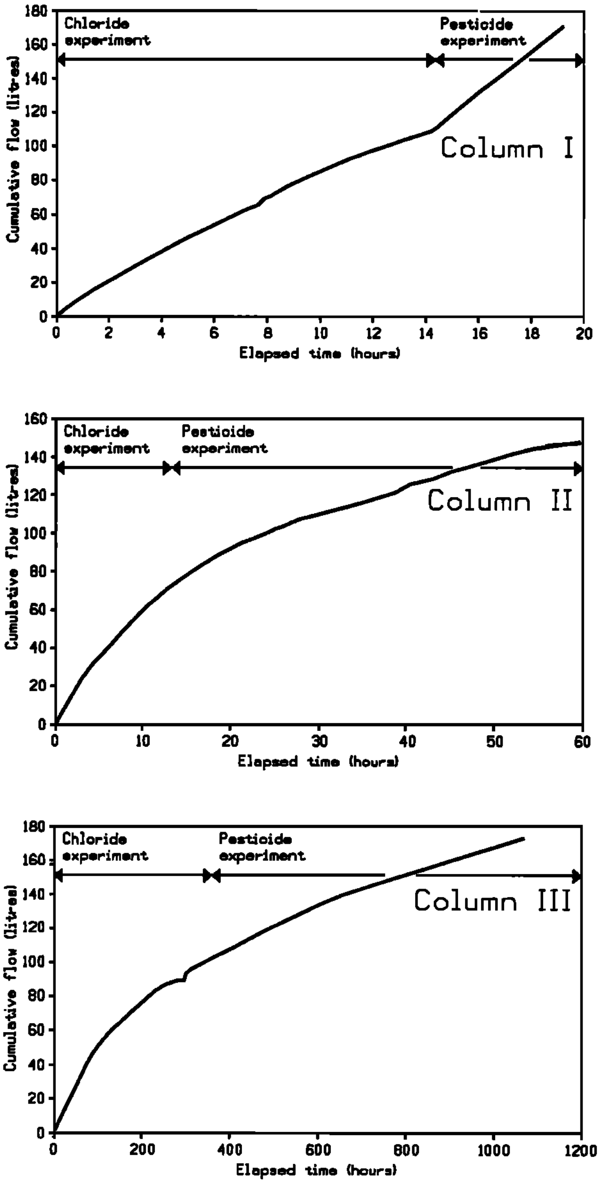

Figure 4. Cumulative flow measured from the undisturbed columns during the tracer and pesticide leaching experiments. All experiments were conducted at unit hydraulic gradient.

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tenuation observed in column III is believed to be due to the longer residence time of the pesticides, which allowed for greater diffusion into the matrix and greater adsorption to the soil solids.

After the tracer experiments were completed the columns were dismantled and the distribution of pesticides within the fractures and matrix were measured to evaluate effects of diffusion and adsorption. Concentrations of pesticides (in μg per kg of dry soil) in the matrix near a fracture in column III are shown in Figure 7a. Concentration of MCPP increases with distance from the fracture wall. This is the type of profile expected to occur for a nonreactive solute or a solute that experiences equilibrium sorption/desorption, because during the flushing period of the experiment, concentrations in the fracture were less than those in the matrix, so solutes in the matrix would diffuse back towards the fracture. The concentration profile for simazine (Figure 7a) shows the opposite

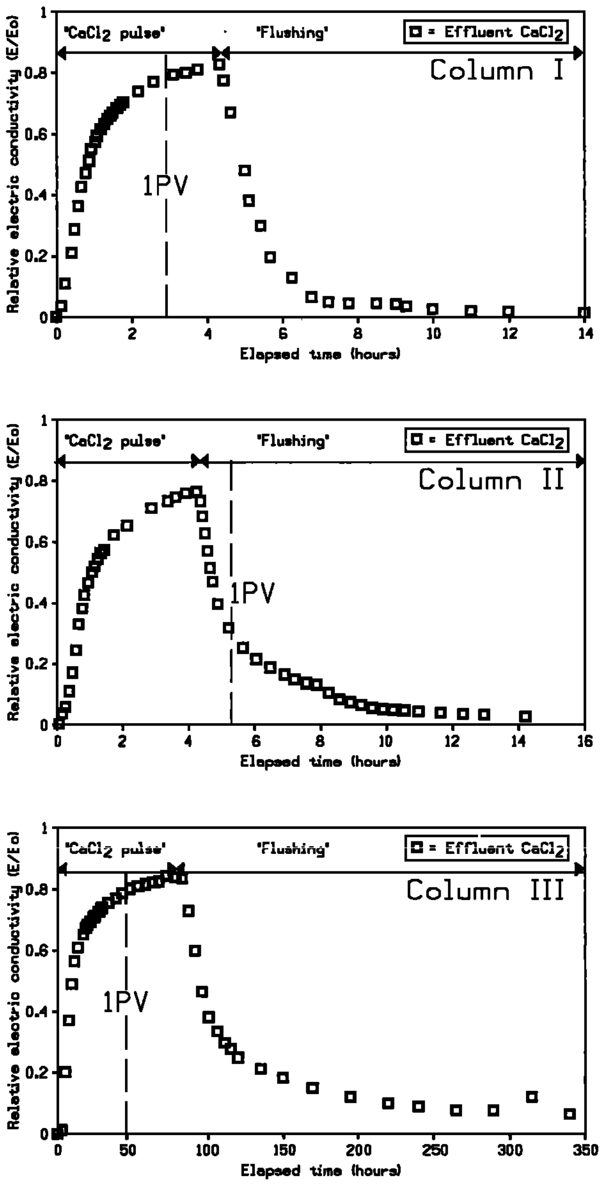

Figure 5. Breakthrough of chloride (measured as electric conductivity) in effluent from large undisturbed columns at unit gradient. Note the difference in x axis scale for each plot.

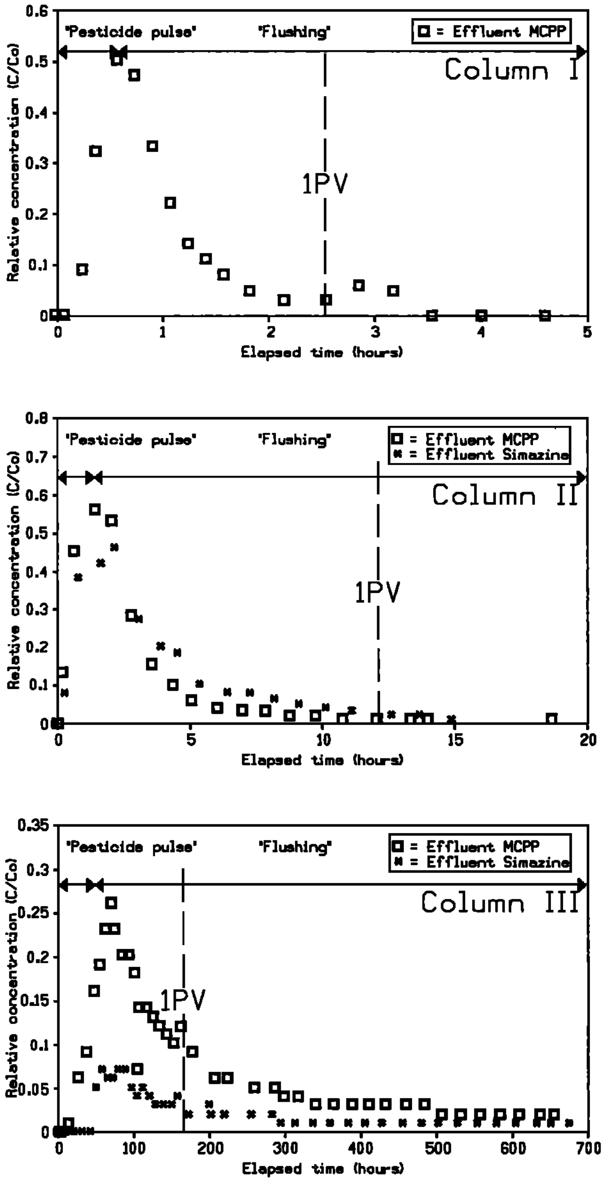

Figure 6. Breakthrough of the herbicides mecoprop (MCPP) and simazine at unit gradient.

trend, with higher concentrations occurring near the fracture. The simazine apparently diffused into the matrix during the pulse injection and sorbed to the matrix solids. When concentrations in the fracture were reduced due to flushing, diffusion of simazine out of the matrix and back into the fracture was delayed or did not occur. This suggests that simazine sorption/desorption is not an equilibrium process and that possibly some of the simazine sorption is irreversible. The MCPP could also have been influenced by nonequilibrium sorption, but to a lesser degree than simazine.

The vertical distribution of the two pesticides with distance along a fracture in column III is shown in Figure 7b. As with the lateral profiles (Figure 7a), the two pesticides exhibit different behavior. While MCPP shows little or no concentration change along the flow path, the simazine concentration decreases with transport distance. This suggests a stronger and/or irreversible adsorption of simazine to the fracture walls. Analyses of the distribution of the pesticides in the shallower col-

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Table 3. Mass Balance of Pesticide Percolation in Column Experiments

		Infiltrated Volumes, L		Pesticide Influent Concentration, mg/L		Pesticides Leached,* % of Infiltrated Amount	
Column	Depth, m	Pesticide Pulse	Flushing Pulse	MCPP	Simazine	MCPP	Simazine
I	1.0–1.5	6	55.9	54.7	···	91	···
II	2.0–2.5	6	55.9	57.6	4.8	91	93
III	4.0–4.5	6	55.9	57.6	4.8	75	21

*Pesticides leached = (Σ mass in effluent/Σ mass in influent) × 100%.

umns (I and II) revealed a more scattered pattern, probably because of the closer and more irregular fracture spacings in the oxidized zone.

4.3. Dye Tracer Experiment

The spacing of conductive fractures and distribution of flow between fractures/macropores and matrix was determined by infiltrating the columns with a fluorescent dye tracer and dismantling the columns under UV light (Plate 1). The fluorescent stain patterns coincided with the oxidized fractures observed under natural light. Approximately 80–100% of the fracture surfaces contained fluorescent staining. In column I, which was collected at 1–1.5 m depth, there was also dye along the few root channels in the column and along soil ped surfaces. The dye tracer was not found along the rubber membrane except for a few spots 1–3 cm in diameter, indicating that there was no leakage around the outside of the sample.

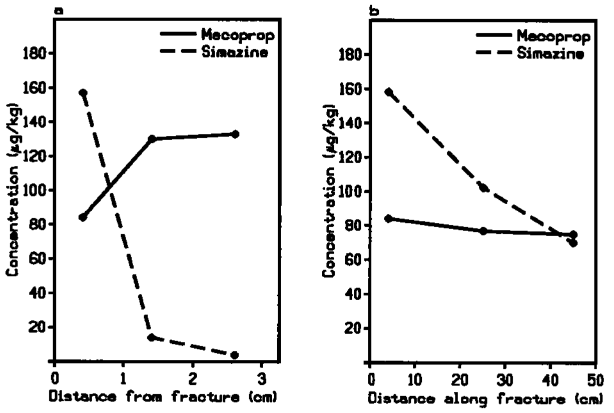

Figure 7. Distribution of MCPP and simazine along a fracture wall and in the matrix near a fracture in column III after flushing of pesticide pulse. (a) Typical profile of concentration versus distance from fracture. (b) Pesticide concentration along fracture wall versus depth.

5. Modeling of Transport in Fractured Media

5.1. Transport Model

The column experiments were simulated using a 2-D model (FRACTRAN) developed for saturated steady state groundwater flow and solute transport in porous or discretely fractured porous formations [*Sudicky and McLaren*, 1992]. The model determines flow rates in the matrix using Darcy's law, based on the hydraulic conductivity (K), the hydraulic gradient (i), and the matrix porosity (θ_m). Fractures are represented as uniform openings between parallel plates and flow velocities in

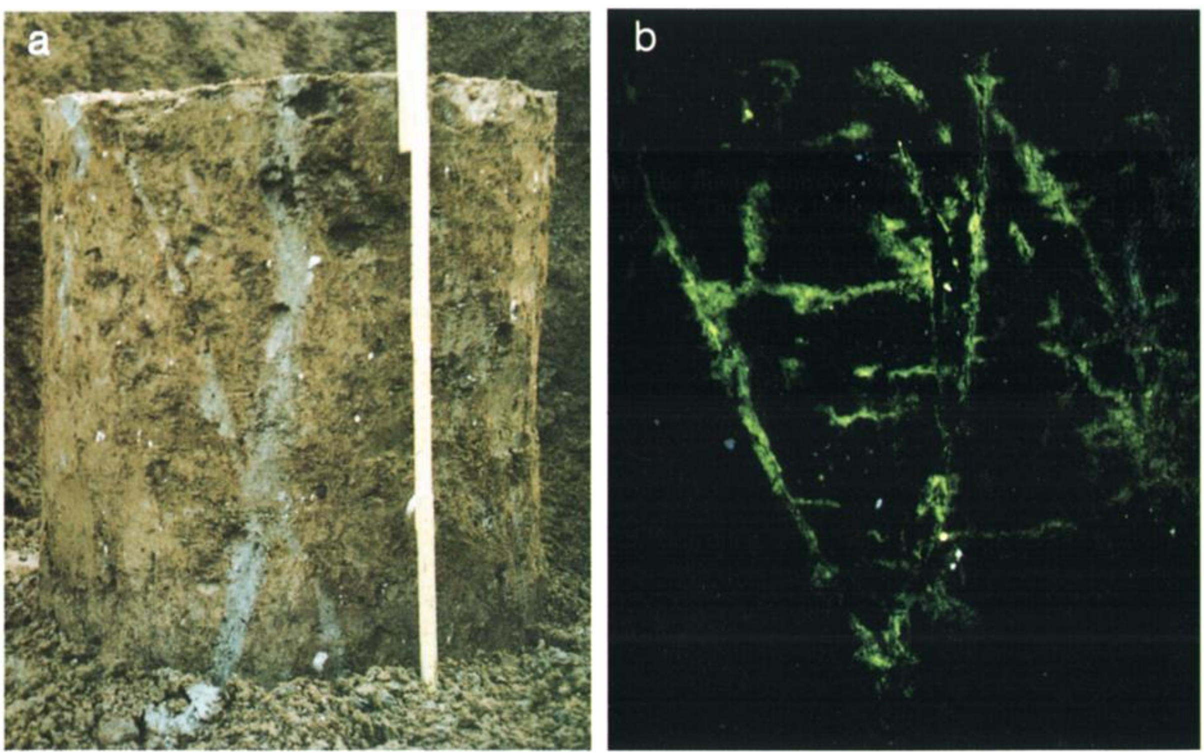

Plate 1. The appearance of column II before and after the fluorescent dye experiment in (a) daylight at sampling and (b) ultraviolet light after dye tracer experiment. The latter image shows the preferential flow paths in the fractures observed during sampling of the column.

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Table 4. Base Case Values of Solute and Porous Media Material Properties Used for EPM and DFMD Modeling of Column Experiments

Parameter	Value[a]	Unit	Source
	Flow System		
Matrix hydraulic conductivity, K	5×10^{-10}	m/s	*Foged and Wille* [1992]
	(10^{-9}–10^{-11})	m/s	
Matrix porosity, θ_m	0.32–0.25	···	measured; Table 1
Matrix dispersivity (longitudinal/transverse), α_l/α_t	0.1/0.01	m	assumed
Fracture dispersivity (longitudinal), α_l^*	0.1	m	assumed
	Effective/Free Water Diffusion Coefficient D^*/D_d		
$CaCl_2$	$6.0 \times 10^{-10}/1.4 \times 10^{-9}$	m^2/s	*Li and Gregory* [1974], *McKay et al.* [1993b]
Mecoprop and simazine[b]	9.7×10^{-11}	m^2/s	estimated[c]
	First-Order Degradation Constant, λ		
Mecoprop and simazine	0	h^{-1}	assumed
	Retardation Factor,[d] R		
$CaCl_2$	1	–	assumed
Mecoprop	1–6.8	–	fitted
Simazine	2–12.5	–	fitted

[a]Range of measured values given in parentheses, where available.
[b]$D^* = D_d \cdot \tau$, where τ is assumed equal to the matrix porosity of column 1 (Table 1).
[c]Diffusion coefficients, D_d, of the pesticides were estimated from diffusion data of aqueous benzoic acid which has a chemical structure comparable to MCPP [*Noulty and Leaist*, 1987; *Worch*, 1993].
[d]R_{matrix} and $R_{\text{fracture wall}}$ are assumed equal values.

the fractures are determined with the cubic law. Fracture spacing ($2B$) and aperture ($2b$) values are specified as part of the nodal input and may be varied in orthogonal network patterns in the 2-D framework. If no fractures are defined in the input data, the model behaves like a conventional porous media transport model. FRACTRAN assigns advective-dispersive equations to both the fractures and the matrix and allows transfer of solutes between fractures and matrix. Coupling between equations describing transport in the fractures and the matrix is provided by the continuity in concentration at the fracture matrix interface and by equality of the normal component of the solute mass flux across this interface. The governing flow and transport equations for the FRACTRAN model are given by *Sudicky and McLaren* [1992].

The model assumes that sorption is a linear equilibrium process independent of sorption capacity of the medium, flow velocity, or residence time. Sorption is described by use of retardation factors for the fractures (R_f) and for the matrix (R). These are required input parameters together with values for effective diffusion coefficient (D^*) for solutes in the matrix and free water diffusion coefficient (D_d). Loss of degradable solutes is described by a first-order decay constant (λ). Hydrodynamic dispersion in the fractures is simulated using the free water diffusion coefficient and an input value of longitudinal dispersivity for the fractures (α_l^*). Input values of longitudinal and transverse dispersivity (α_l and α_t) are also needed to determine hydrodynamic dispersion in the matrix. The input parameters for the column modeling (Table 4) use the same symbols as used by *Sudicky and McLaren* [1992].

Table 5. Calculated $CaCl_2$ Transport in Columns Based on the EPM Approach, Using Total Porosity Values

		Velocity of Chloride Front ($C/C_0 = 0.5$), m/d	
Column	Depth, m	Calculated With EPM (θ_{total})	Measured
I	1.0–1.5	3.3	10.9
II	2.0–2.5	1.9	13.8
III	4.0–4.5	0.2	1.18

5.2. EPM Total-Porosity Simulations of Chloride Transport

The EPM plug flow velocity or the average velocity at $C/C_0 = 0.5$ of the advective front of a nonreactive tracer was evaluated using

$$v_s = \frac{Ki}{\theta_e} \tag{2}$$

where K is the bulk hydraulic conductivity, i is the hydraulic gradient, and θ_e is the effective porosity (set equal to total porosity for the following simulations).

Using (2) the EPM plug flow velocity, v_s, was calculated for each column and compared with the velocity of the nonreactive chloride tracer front at $C/C_0 = 0.5$ (Table 5). For all three columns tested the calculated EPM plug flow transport velocity is 3–8 times slower than the measured velocity of the chloride at $C/C_0 = 0.5$. This means that for the applied hydraulic gradient of 1, the EPM plug flow transport (using the total porosity as effective porosity) would underestimate the rate of tracer migration.

Simulations of chloride transport in each column were then carried out using the FRACTRAN model in EPM mode with values of v_s determined from (2) and other parameters as shown in Table 4. The hydraulic conductivity value used to determine v_s for each column was determined from the hydraulic conductivity during the injection portion of each tracer experiment (Table 2). Actual flow rates and hydraulic conductivity values varied as shown on Figure 4, but simulations using

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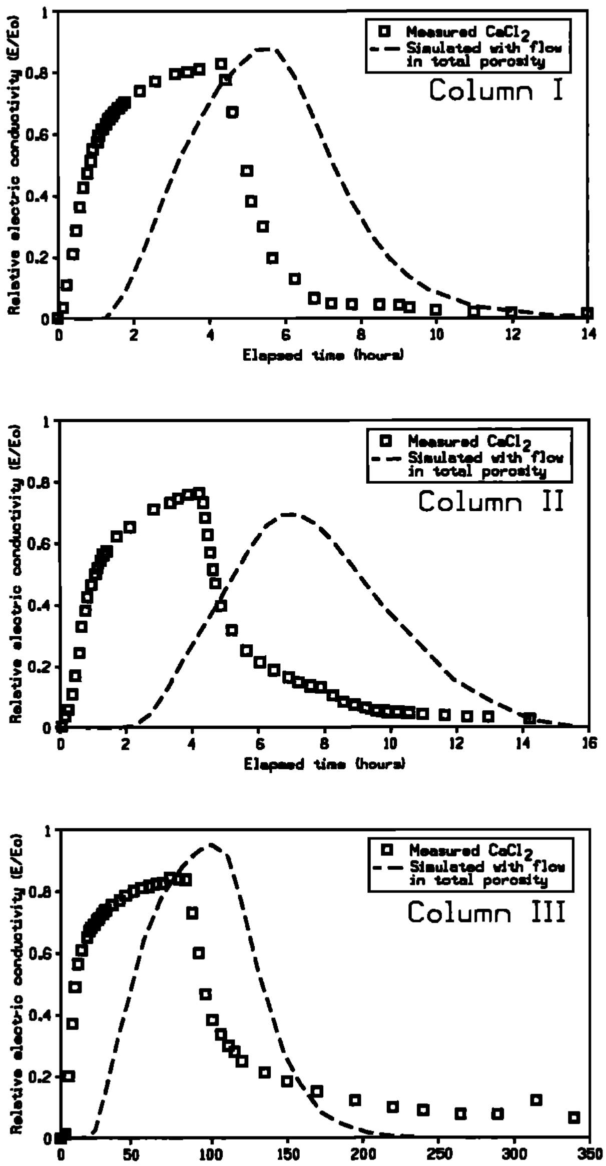

Figure 8. Simulated breakthrough of chloride compared to measured values in effluent from column experiments. Simulations were based on an EPM transport model (using the total porosity as effective porosity).

a stepwise change in hydraulic conductivity showed this had little influence on the breakthrough curve. As expected, the simulated breakthrough curves (Figure 8) are retarded relative to the measured curves and also show much greater spreading and lower peak concentrations (which are influenced by the assumed dispersion coefficients).

It is expected that at a lower hydraulic gradient and hence lower flow rate, the EPM modeling approach using total porosity would more closely fit measured breakthrough curves. At lower gradients or flow rates there is more time available for solutes to diffuse throughout the matrix and "sample" more of the total porosity. Under natural conditions, however, situations of high gradient and rapid fracture flow can occur during rainstorms or seasonally wet periods and much of the transport may occur during these periods. At the field site where the clayey till columns were collected vertical hydraulic gradients of 0.6–1 have been measured during wet periods [*Jørgensen and Spliid*, 1998].

5.3. Hydraulic Apertures and Fracture Porosity

The discrete fracture approach simulations of flow and transport through the columns were based on the primary assumption that flow through individual fractures can be represented by an analogy to laminar flow between two smooth parallel plates [*Snow*, 1968, 1969]. In this case the hydraulic conductivity of a fracture, K_f, is given by the so-called cubic law

$$K_f = \frac{(2b)^2 \rho g}{(12\mu)} \tag{3}$$

where

$2b$ fracture aperture;
ρ fluid density;
g acceleration of gravity;
μ dynamic viscosity of water.

For steady state flow the average velocity in the fracture, v_f, is given by

$$v_f = K_f i \tag{4}$$

where i is the hydraulic gradient along the fracture.

It is also necessary to make assumptions about which of the fractures contributes to flow. The widespread distribution of fluorescent dye in the fractures of the samples after the dye flushing indicates that flow is occurring in many fractures rather than in just one or two. However, in the field some widely spaced fractures were more visually prominent and may have large aperture values. Two conceptual scenarios (shown in Figure 9) were considered for the modeling: (1) all fractures "closed" except for one vertical fracture through the center of the sample and (2) all fractures "open" with the system represented by an orthogonal network of vertical, continuous, equal-aperture fractures with spacing equal to the average distance between vertical fractures measured in each column.

For each column, values of "hydraulic aperture," which is the aperture value calculated using the "cubic equation" and the measured value of vertical hydraulic conductivity K_z, were determined. These values become input data for the subsequent discrete fracture transport simulations. For the case of a single fracture through the sample (scenario 1)

$$K_z = \frac{K_f a_f}{a_s} + K_m \tag{5}$$

where K_m is the matrix permeability; a_f is the cross-sectional area of fracture, equal to $2bd$, where d is the diameter of sample; and a_s is the cross-sectional area of sample, and hence

$$2b = \left(\frac{(K_z - K_m) a_s \times 12\mu}{d \rho g} \right)^{1/3} \tag{6}$$

For the case with a set of orthogonal vertical fractures (scenario 2),

$$K_z = \frac{2(2b) K_f}{2B} + K_m \tag{7}$$

and

$$2b = \left(\frac{(K_z - K_m) 2B \times 6\mu}{\rho g} \right)^{1/3} \tag{8}$$

where $2B$ equals the fracture spacing.

Fracture porosity for scenario 2 is given by

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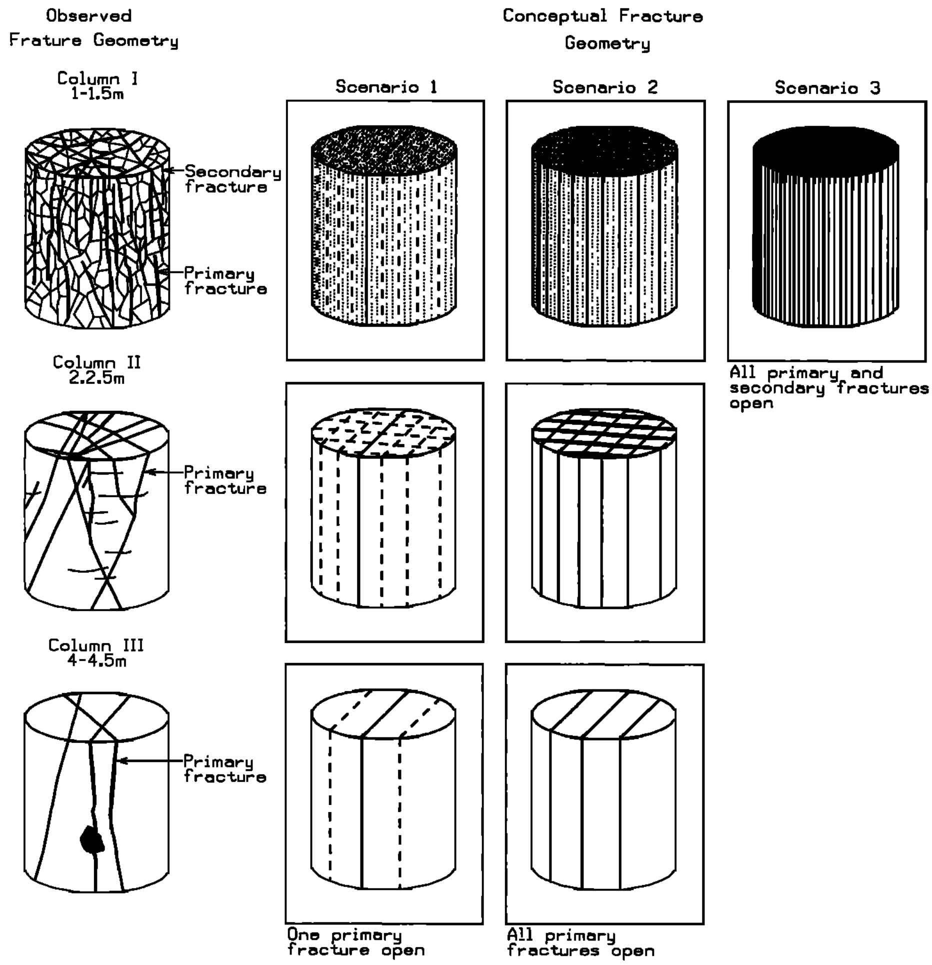

Figure 9. Conceptual scenarios of fracture geometry used in the numerical flow and solute transport simulations versus observed fracture geometries. The model simulates transport in a 2-D vertical slice through the column.

$$\theta_f = \frac{2(2b)}{2B} \qquad (9)$$

Calculated hydraulic fracture apertures for the cases with single and multiple primary fractures (scenarios 1 and 2 in Figure 9) are shown for the three columns in Table 6. The calculations used mean hydraulic conductivity values measured in the injection portion of the chloride tracer experiments (Table 2) and the fracture spacings measured by the fluorescence dye tracer test (Table 1). The resulting hydraulic aperture values for all columns range from 69 to 199 μm for the single fracture case and from 42 to 84 μm for the multiple fracture case. Values of aperture determined for the pesticide injections were slightly different, because of variations in the flow rate and calculated hydraulic conductivity, K_z. However, because aperture is related to the cube root of K_z, as shown in (6) and (8), the influence was small.

5.4. Discrete Fracture Simulations of Chloride Transport

To evaluate the influence of transport through a single fracture or through multiple fractures, the chloride tracer breakthrough was simulated for each column (conceptual scenarios 1 and 2 in Figure 9) using FRACTRAN with discrete fractures in a low–hydraulic conductivity matrix (DFMD approach).

Table 6. Hydraulic Apertures and Flow Velocity in Fractures Calculated From Hydraulic Conductivity and Fracture Spacing

Column	Depth, m	Fracture Spacing, m	Vertical Hydraulic Conductivity, m/s	Hydraulic Aperture, μm (Flow Velocity, m/d): Single Fracture Open	Hydraulic Aperture, μm (Flow Velocity, m/d): All Primary Fractures Open	Calculated Fracture Porosity for Case With All Primary Fractures Open
I	1.0–1.5	0.06 primary; 0.025 secondary	1.2×10^{-5}	199 (2137)	84 (381)	0.0028
II	2.0–2.5	0.10	6.9×10^{-6}	166 (1487)	83 (372)	0.0016
III	4.0–4.5	0.16	6.1×10^{-7}	69 (257)	42 (95)	0.00053

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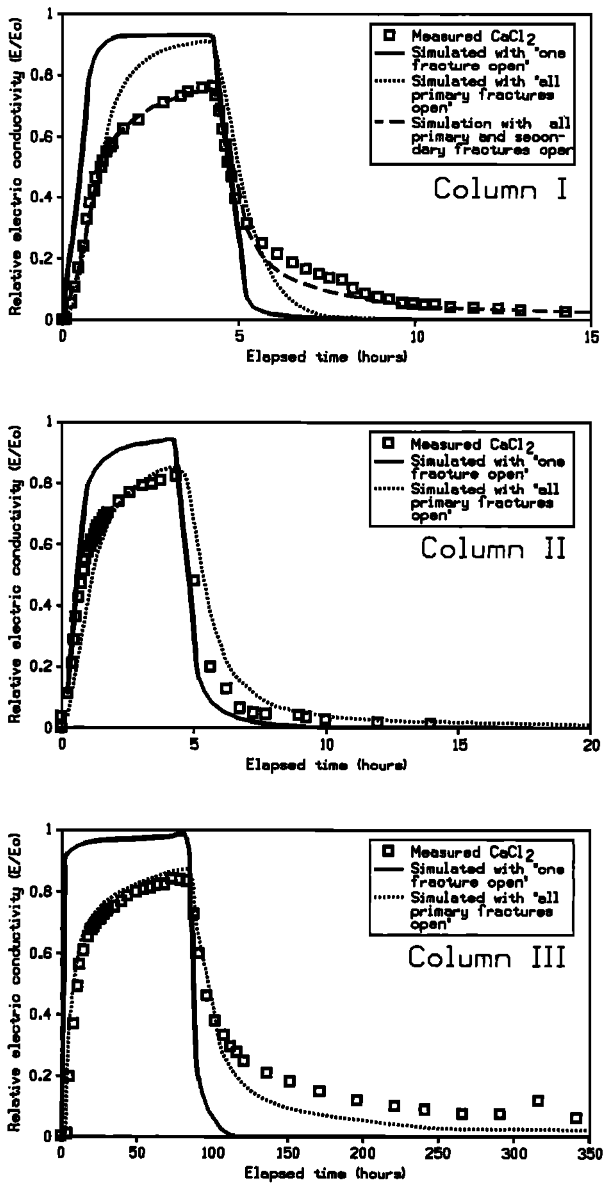

Figure 10. Simulated breakthrough of chloride compared to measured values in effluent from column experiments. Simulations are shown for the conceptual scenarios of fracture flow shown in Figure 9.

The simulations were carried out by using the fracture spacings and the hydraulic aperture values calculated by the cubic equation for each of the scenarios (Table 6) along with the base case parameters shown in Table 4. The simulated breakthrough curves are shown together with the experimental breakthrough curves (BTCs) in Figure 10.

The simulations for scenario 1, with only one primary fracture "open," clearly overestimate breakthrough rates and concentrations in all three columns. The simulations for scenario 2, with all fractures open, provided a reasonable approximation to the chloride BTC data for columns II and III, but not for column I. Hence for columns II and III the fracture transport aperture values (satisfying the BTCs) are very similar to the hydraulic aperture (satisfying flow measurements) when using the multiple fracture approach for the columns II and III. This result suggests that flow must occur through many fractures rather than just one or two large fractures and furthermore indicates that flow is distributed along the fracture planes rather than directed into a few larger aperture channels within the individual fractures. These findings are consistent with the results of the dye tracer test (Plate 1), which showed dye tracer along most of the fractures.

For column I both the cases with a single fracture open (199 μm) and with all primary fractures (84 μm) open overestimated the chloride breakthrough. Fitting the simulations to the breakthrough data by reducing the fracture aperture resulted in a significant underestimation of the flow rate (not shown). Hence the flow system of column I appears to not be adequately represented by either scenario of flow through the primary fractures (Figure 9). A new set of fractures (scenario 3) was introduced for additional column I simulations. These secondary fractures represent the soil ped interfaces, which were found only in this shallow column. By including flow and solute transport along the soil ped interfaces, the simulations were fit to the BTC by reducing the aperture of the primary fractures and increasing aperture of the secondary fractures, while maintaining a constant vertical hydraulic conductivity for the column. A good fit to the chloride BTC was achieved using apertures of 78 and 27 μm for primary and secondary fractures, respectively (Figure 9). For these aperture values the calculated hydraulic conductivity still equals the measured value for the column, so the simulations fit both the flow and the transport data.

The simulations show that matrix diffusion is sufficient to cause the large difference between the average chloride velocity (1.18–10.9 m/d at $C/C_0 = 0.5$) and the calculated fracture flow velocity (95–281 m/d). The simulations also confirm that the long chloride "tails" observed during the flushing portions of the experiments are due to diffusion from the matrix pores back into the fractures and do not imply that there was significant sorption of the chloride.

6. Pesticide Transport Simulations

The simulations of the pesticide experiments were carried out with the FRACTRAN model using the same fracture distribution and aperture values as for the "best fit" simulations of chloride transport (Table 6) and the base case parameters for the respective pesticides, as shown in Table 4. The retardation factors were the only variables used for fitting simulations to the pesticide data and R in the matrix was assumed equal to R_f in the fractures. Pesticide losses due to decay were assumed to be zero. Sensitivity analyses by FRACTRAN using decay rates measured for MCPP in the till, and literature values for simazine indicated that decay would be negligible over the short residence times measured in the column experiments [*Jørgensen and Spliid*, 1998]. Experimental and simulated breakthrough curves for the pesticides are shown in Figure 11.

The simulated MCPP concentrations for column I closely fit the measured values using a retardation factor (R) of 1. In column II the simulated MCPP concentrations also provided a reasonable fit to the data using an R value of 1. The simulated simazine concentrations for column II fit the data using an R value of 2. A good fit was achieved for MCPP using an R value of 6.8 in column III. For simazine in column III an R value of 12.5 was required to fit the first arrival of the pesticide, but for the rest of the BTC the simulated values were much greater than the measured values. For the best fit simulation in column III the actual mass recovery for simazine is only approximately

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40% of the simulated value. Subsequent simulations showed that it was not possible to fit the entire simazine breakthrough curve in column III simply by varying the R value.

The large difference in fitted R values between columns I and II and column III is likely due to nonequilibrium sorption of the pesticides, especially simazine. In columns I and II flow rates were a factor of 20–125 times higher than for column III (Table 2), and hence there was more time for sorption of the pesticides in column III. This is consistent with the profile of simazine in the matrix of column III (Figure 7a), which showed that diffusion of simazine out of the matrix during the flushing stage of the experiment was much slower than for MCPP. As well, the lower values of mass recovery (Table 3) observed for both pesticides in column III are also consistent with greater sorption (or possibly some degree of irreversible sorption) in column III. Time-dependent sorption of various organic contaminants has been observed in numerous studies, often with about half of the sorption occurring over time periods of a few minutes or hours and with the remainder occurring over periods of days or months [e.g., *Lee et al.*, 1988; *Brusseau and Rao*, 1990]. As well, batch studies by *Weber* [1966] and *Bailey et al.* [1968] have shown significant and partially irreversible adsorption of organic pesticides like simazine and atrazine.

The influence of nonequilibrium sorption is often difficult to distinguish from that of matrix diffusion in fractured soils, since both result in attenuation of solute concentrations. However, the DFMD approach explicitly accounts for attenuation due to matrix diffusion, so the difference in fitted R values for the pesticides in the three columns is a strong indicator of nonequilibrium sorption. Differences in mineralogy or organic content of the till may also be partly responsible for the differences in fitted R values. However, all the columns were collected from the same deposit and, particularly for columns II and III, which were both taken below the root zone, large differences in sorption characteristics were not expected. Matrix retardation factors for MCPP in the columns were calculated based on equilibrium distribution coefficient (K_d) values (0.27–0.86) reported for similar Danish soils [*Helweg*, 1996; G. Felding, personal communication, 1994] using

$$R = 1 + K_d \frac{\rho_b}{\theta} \tag{10}$$

where ρ_b is the bulk density of the matrix [*Freeze and Cherry*, 1979]. The R values for MCPP were calculated on the basis of the mean values of bulk density and porosity for all the columns and ranged from 2.4 to 6.8. These estimated R values are larger than the fitted values (equal to 1) obtained from the MCPP tracer experiments in columns I and II, while the fitted value of 6.8 obtained for column III is within the range of the estimated values. This suggests that for the relatively fast flow rates in columns I and II there was inadequate time for MCPP sorption to occur, but for the slower flow rate observed in column III, sorption more closely approached what was observed in the batch sorption experiments.

The calibrated DFMD model was used for additional simulations (not shown) assuming a constant concentration source of MCPP, an R value of 6.8, and all other values as given in Table 4. On the basis of these simulations the breakthrough arrival velocities (at $C/C_0 = 0.5$) range from 0.09 to 21.1 m/d in columns I–III. Corresponding velocities calculated for an EPM approach using total porosity at the same infiltration rates and the same R value (6.8) range from 0.008 to 0.32 m/d. The ratio of the simulated MCPP velocities determined from the DFMD and EPM approaches ranges from approximately 10 to 65. By comparison, the ratio of measured chloride velocity (also at $C/C_0 = 0.5$) from the tracer experiments and the simulated EPM velocity (at $R = 1$, as shown in Table 5) ranges from 3 to 8. This indicates that the error in calculated contaminant velocity caused by assuming EPM conditions increases as the R value of the contaminant increases. This is because in the DFMD simulations the contaminant is in contact with a smaller portion of the matrix and hence has less opportunity for sorption. As fracture spacing or flow velocity decreases, the difference between the EPM and the DFMD approach also decreases, but as indicated by this experimental study there are likely many geologic settings where an assumption of EPM behavior is not valid.

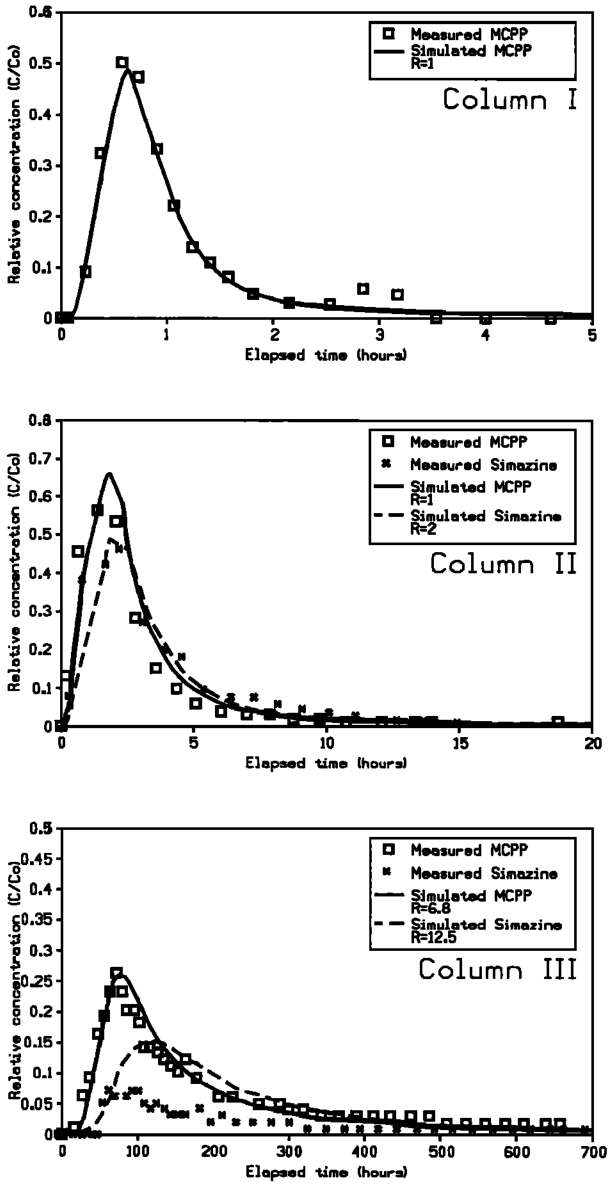

Figure 11. Simulated breakthrough of pesticide pulse compared to measured values in effluent from column experiments.

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7. Conclusions and Implications

Groundwater movement through the undisturbed columns in this study is controlled by flow through the visible stained fractures and, in the case of the uppermost column, by flow through a few root channels. The saturated hydraulic conductivity of the columns was similar to field-measured values in the tills, which decrease with depth and fracture density, indicating that the columns were generally representative of field conditions.

Results of transport experiments using nonreactive and reactive solutes are consistent with a conceptual model of advective transport through the fractures combined with diffusion into the relatively immobile pore water in the matrix between the fractures. This was indicated by the characteristics of the solute breakthrough curves (rapid initial concentration rise, followed by slower increase during the injection and a very long "tail" during the flush), by the measured distribution of solutes in the matrix adjacent to the fractures, and by the distribution of a dye tracer along the fractures. At high flow rates the reactive pesticide tracers migrated essentially as nonreactive tracers, apparently because of nonequilibrium sorption, but these same tracers were significantly retarded at the slowest flow rate.

Simulations with an EPM model assuming transport through the total porosity of the material greatly underestimated solute transport rates at high hydraulic gradients as represented by the experiments. Although the simulations could likely have been fit to the lab data by decreasing the effective porosity (θ_e), this approach is expected to be less suitable for field applications because θ_e as well as the dispersion coefficient would change with physical scale, elapsed time, and flow rate. Simulations with a discrete fracture flow/matrix diffusion (DFMD) model were capable of fitting both the hydraulic data and the solute transport data in all columns using measured fracture spacings and calculated hydraulic aperture values of 27–84 μm. This method, which is based on numerical descriptions of the actual interaction of the controlling processes for transport, is expected to better incorporate changes in transport behavior due to different scale, time, and flow rate than the EPM approach but also requires more detailed information on the physical characteristics of the deposits. The study suggests that numerical models based on discrete fracture/matrix diffusion have potential for application in assessment of pesticide leaching risk in fractured clay-rich deposits and may have significant advantages over conventional EPM methods, especially in cases where fractures are widely spaced. The hydraulic aperture values determined from this study are similar to values determined at other sites in glacial clays in northern Europe and North America, so for risk assessment purposes it may be possible to use statistically determined fracture aperture distributions. Fracture spacing and depth will be key parameters needed for field applications and this could be very difficult to determine in cases where hydraulically conductive fractures do not have distinctive oxidation/reduction staining.

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